Enhancement of scale factor control accuracy of modulation algorithm for a closed-loop fiber-optic gyroscope.

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Abstract

TODO abstract

1 Motivation

iFOG principle of operation is based on the Sagnac effect. Its design has already been excercised for many years. However, as technology evolves, the limits holding the development back are pushed forward, enabling for better precision, miniaturized size or lower power consumption. At the same time, the ever progressing industry finds new and more strict applications for the gyroscopes, setting the requirements bar even higher. Such context makes the already existing designs being tested and optimized in every possible aspect. This applies equally to all the optical and electro–optic componenents, the electronics and the driving software. Crucial part of the control system of such gyroscope is the modulator. In a closed-loop design it is a vital element of the whole feedback loop, which directly affects the overall performance. It drives the optical circuit into a predefined state, in which the circuit response can be demodulated and error signals determined. Modulator accuracy has critical impact on the resulting performance. This paper takes a closer look at the all–digital implementation of the modulator subsystem and particularily on the algorithm controlling it. Mathematical analysis is conducted to allow for pure software enhancements in the algorithm accuracy. Then, by placing the algorithm in the context of various hardware configurations, its accuracy is also determined for each of them.

2 Principle of operation - open–loop configuration

A fibre–optic gyroscope utilizes Sagnac effect. It generates a light beam, which is ideally split in 50/50 ratio. Each half then enters a loop of relatively long fiber through each of its ends and propagates towards the other end. The fiber length L determines light transit time τ through it (Equation 1, where n is the refraction coefficient, affecting light speed in the medium).

$$\tau = \frac{L}{n \cdot c} \tag{1}$$

The halves go past each other and exit on the opposite ends of the fiber. The optical circuit is arranged, so that the two exitting beams interfere with each other and the resulting light intensity is measured at a detector. When the circuit stays at rest, the beams interfere with zero phase shift ($\Delta\phi$), resulting in the maximum intensity P_0 measured at the detector. However, when the circuit rotates, the two halves travel paths of slightly different lengths, resulting in a non–zero phase shift. The halves relative phase shift at the interference point is rotation dependent and in this paper it is reffered to as the Sagnac–induced phase shift:

$$\Delta\phi_s = \frac{L \cdot D \cdot \omega}{n \cdot c \cdot \lambda} \tag{2}$$

where D - fiber loop diameter, ω - angular velocity, λ - wavelength. Figure 1 depicts this phenomenon. Both halves enter the loop at the entry point. After the transit time they exit through the same point. However, because of non–zero rotation, this Entry/Exit point moves, effectively shrinking the path of one half and extending the path of the other. Path difference is relatively small, thus it is expressed as phase difference relative to the wavelength, rather than absolute length in meters.

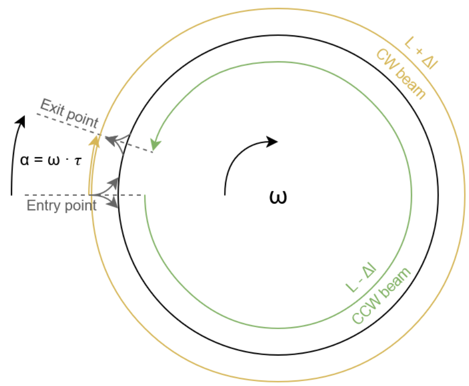


Figure 1: Two counterpropagating light beams enter the fiber loop at the *Entry point*. Their phase is initially in sync. After the transit time τ they exit and interfere. However, during that time, the *Entry/Exit point* moves according to rotation rate ω (by the angle $\alpha = \omega \cdot \tau$) and the CW and CCW beams covered non–equal path lengths ($L \pm \Delta l$). As a result they differ in phase.

The formula in Equation 3 and Figure 2 is the optical circuit response. It describes how the measured light intensity depends on the total phase shift.

$$P(\Delta\phi) = \frac{P_0}{2} \cdot (1 + cos(\Delta\phi)) \tag{3}$$

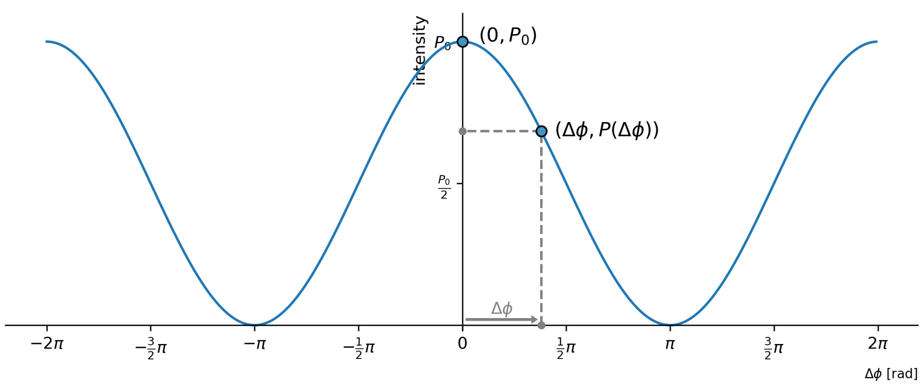


Figure 2: Optical circuit response function is a shifted cosine. The two exemplary operating points refer to the circuit being at rest, yielding the maximum possible intensity - in $(0, P_0)$ - and subjected to rotation, at non–zero phase shift and decreased intensity - in $(\Delta\phi, P(\Delta\phi))$.

When the circuit is at rest ($\omega = 0 \rightarrow \Delta\phi = 0$), the operating point resides at peak intensity P_0. A non–zero rotation results in a phase shift which moves the operating point according to rotation rate and direction, decreasing the measured intensity. Two problems are linked with such open–loop operation. Firstly, the intensity drop does not allow to tell the direction of rotation, as the response function is symmetrical. Secondly, in the resting region, the slope is flat, thus the sensitivity is zero. With greater rotation rate, the slope becomes steep, reaching its maximum sensitivity in $\pm\frac{\pi}{2}$, but then decreases again. For these reasons a closed–loop configuration will be superior in performance.

3 Modulation scheme - closed–loop configuration

In closed–loop configuration of an iFOG, the control system biases optical circuit sequentially in four predefined points - $\Delta\phi \in \{+\frac{3}{2}\pi, -\frac{1}{2}\pi, -\frac{3}{2}\pi, +\frac{1}{2}\pi\}$ - utilizing the *four–step modulation* (Figure 3). This allows for linearizing the response, keeping the circuit at maximum sensitivity and detecting the rotation direction.

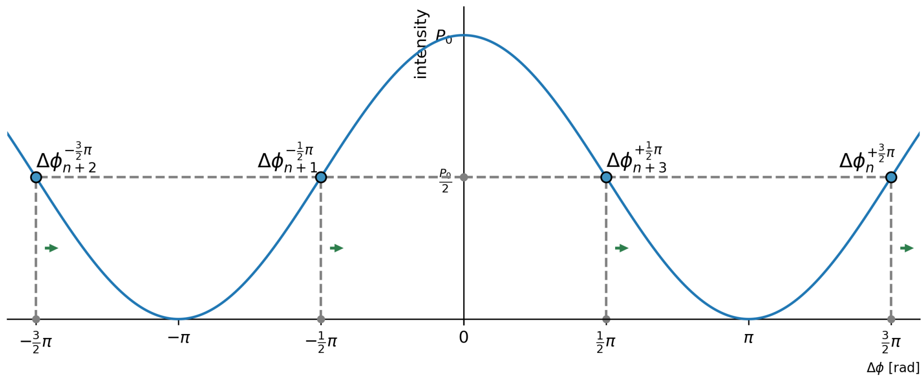


Figure 3: Circuit biased in four points of highest sensitivity.

Compared to *two–step modulation* this scheme is supreme, as it allows for controlling the scale factor. This is crucial due to several reasons. Modulator alters the phase of passing light through an electro–optic device, placed within the fiber loop. The applied phase shift is controlled by an electronic circuit. It performs digital calculations and applies a corresponding voltage using a digital–to–ananlog converter and an electronic front–end. Then the voltage is translated to a phase shift value by the electro–optic actuator. All of these elements are subjects temperature drift and aging, which affect the total combined gain. For this reason, constant calibration is required for maintaining measurement accuracy. In short, the control algorithm needs to know at all times where - voltage wise - are the desired bias points. Diagrams in Figure 4 depict how the biased circuit responds to various conditions - scale factor error and rotation present. From there, the control loop can determine how to adjust the parameters, to place the points back to where they shall ideally reside.

From the point of view of the control algorithm, the scale factor can be reffered to as the half–wave voltage V_π, even though it is not the only parameter affecting the effective scale factor. However, because of the assumed linearity of the other compounds, it is convenient to aggregate all the linear error sources into one parameter (other, non–linear errors, such as DAC INL are out of scope of this analysis).

Device control system biases the circuit continously with the following bias point order:

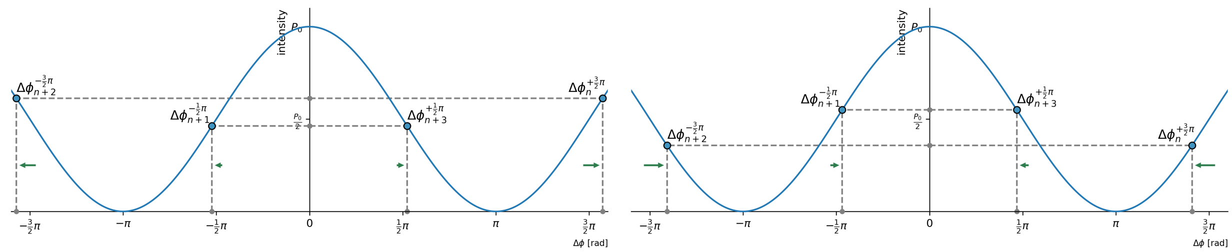
$$\{\Delta\phi_{4k}^{+\frac{3}{2}\pi}, \Delta\phi_{4k+1}^{-\frac{1}{2}\pi}, \Delta\phi_{4k+2}^{-\frac{3}{2}\pi}, \Delta\phi_{4k+3}^{+\frac{1}{2}\pi}\} \tag{4}$$

where the superscripted phase is the intended effective phase shift for the given sample.

The modulator runs with the period of half the transit time τ. This allows for the effective phase shift to be a result of a relative shift between every n'th and $(n-2)$'th samples. For this reason the phase shift applied by the modulator for each sample can be half of the desired amplitude:

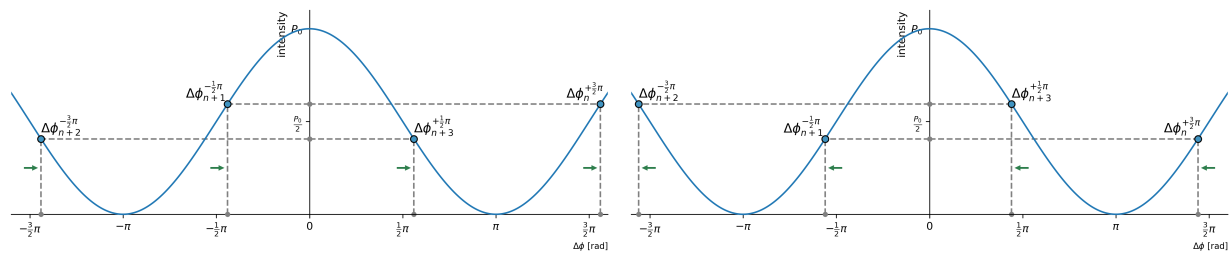
$$\begin{aligned}
\phi_{4k}^{+\frac{3}{2}\pi} &= +\frac{3}{4}\pi \cdot (1 + \delta_{4k}) \\
\phi_{4k+1}^{-\frac{1}{2}\pi} &= -\frac{1}{4}\pi \cdot (1 + \delta_{4k+1}) \\
\phi_{4k+2}^{-\frac{3}{2}\pi} &= -\frac{3}{4}\pi \cdot (1 + \delta_{4k+2}) \\
\phi_{4k+3}^{+\frac{1}{2}\pi} &= +\frac{1}{4}\pi \cdot (1 + \delta_{4k+3})
\end{aligned} \tag{5}$$

where δ_n denotes a close to zero scale factor error for the corresponding sample.



(a) Scale factor too high - bias points are shifted away from zero, resulting in intensity imbalance.

(b) Scale factor too low - bias points shifted towards zero, resulting in opposite imbalance.



(c) Phase shift induced by positive rotation - bias points shifted right. Points $\{n, n+1\}$ correspond to higher intensity than points $\{n+2, n+3\}$.

(d) Phase shift induced by negative rotation - bias points shifted left. Points $\{n, n+1\}$ correspond to lower intensity than points $\{n+2, n+3\}$.

Figure 4: Four–step modulation allows for detection of scale factor error and Sagnac–induced phase shift.

Additionally, to null the effect of the rotation–induced phase shift, the modulator adds an ever–updating ramp signal with the delta of $\Delta\phi_r = -\Delta\phi_s$ - equal in amplitude and opposite in sign. This way the algorithm cancells the Sagnac effect, while knowing how much of it has been cancelled and thus it knows the exact rotation rate. In every cycle, the software monitors error inputs and adjusts control signals accordingly. A detailed operation of the control loop is described in TODO-lefevre.

4 Hardware architecture

4.1 Classic architecture - Dedicated gain stage

4.2 Thin architecture - All-digital approach

5 Algorithm requirements and design

6 Scale factor error

TODO
Scale factor error has two sources. Primarily the algorithm needs to adjust the nummerical value precisely to control the scale factor. Error in calculating this value will directly impact the accuracy. However, even with a perfect algorithm teh quantization noise resulting from the usage of a DAC leaves some error preset at all times. While the first factor is a subject for algorithmic optimization, the second one resuts from the physical circuit impleemntation and depends on several followign parameters.

Firstly, the relative scale factor error can be denoted as in Equation 6.

$$\delta(n) = \frac{1}{2} \cdot \frac{\Delta v}{V_\pi(n)} \tag{6}$$

Voltage resolution of the used DAC results directly from the number of bits of the converter (Equation 7).

$$\Delta v = V_{ref} \cdot 2^{-N} \tag{7}$$

Minimal DAC reference voltage shall to be adjusted to cover all the required operating span (Equation 8).

$$V_{ref}^{min} = V_{\pi}^{max} \cdot R \cdot M \tag{8}$$

where range $R = 3.5$ results directly from the algorithm implementation (it shall be able to drive the values equivalent to $\pm 1\frac{3}{4}\pi$, yielding the span of $R = 3.5$) and margin $M \in [1.0; 1.1]$. When determining the final reference voltage it is advantageous to align it with a voltage multiple of one of typically available chips. It makes implementing the elctronic circuit more straightfroward, thus eliminating possible noise sources. The resulting reference voltage will in most cases be slightly higher then the minimal value (Equation 9). In practice, the best compromise is to utilize this excess for the margin M, so that after adding it, the $V_{ref} \approx V_{ref}^{min}$.

$$V_{ref} = \lceil V_{ref}^{min} \rceil \tag{9}$$

Substituting TODO yields the formula for relative scale factor error in a worst case scenario, when the real sclae factor remains exactly in between the two controllable, discrete phase shift values, but with the assumption, that the software has perfectly calculated the scale (no error introduced by the algorithm).

$$\delta(n) = \frac{1}{2} \cdot \frac{\lceil V_{\pi}^{max} \cdot R \cdot M \rceil \cdot 2^{-N}}{V_{\pi}(n)} \tag{10}$$

An exemplary calculation assuming some real–world values:

$$\begin{aligned} \delta_{ex} &= \frac{1}{2} \cdot \frac{\lceil 4.5V \cdot 3.5 \cdot 1.04 \rceil \cdot 2^{-16}}{3.5V} = \frac{\lceil 16.38V \rceil \cdot 2^{-16}}{7V} \\ &= \frac{\overbrace{16.384V}^{4 \cdot 4.096V} \cdot 2^{-16}}{7V} = 2^{-16} \cdot 2.34 = 2^{-16} \cdot 2^{1.23} \end{aligned} \tag{11}$$

6.1 Scale factor error feedback

$$\Delta\phi_n = (\phi_n - \phi_{n-2}) + \Delta\phi_n^{sagnac} \tag{12}$$

Assuming no Sagnac effect:

$$\begin{aligned} P_n^{+\frac{3}{2}\pi} &= \frac{P_0}{2} \cdot (1 + cos(\Delta\phi_n)) = \frac{P_0}{2} \cdot (1 + cos(\phi_n - \phi_{n-2})) = \\ &= \frac{P_0}{2} \cdot (1 + cos(+\frac{3}{2}\pi \cdot (1 + \frac{\delta_n + \delta_{n-2}}{2}))) \\ &= \frac{P_0}{2} \cdot (1 + \frac{\delta_n + \delta_{n-2}}{2}) \\ P_{n+1}^{-\frac{1}{2}\pi} &= \frac{P_0}{2} \cdot (1 + \frac{\delta_{n+1} + \delta_{n-1}}{2}) \\ P_{n+2}^{-\frac{3}{2}\pi} &= \frac{P_0}{2} \cdot (1 + \frac{\delta_{n+2} + \delta_{n}}{2}) \\ P_{n+3}^{+\frac{1}{2}\pi} &= \frac{P_0}{2} \cdot (1 + \frac{\delta_{n+3} + \delta_{n+1}}{2}) \end{aligned} \tag{13}$$

$$\begin{aligned} P(\phi_B + \delta) &= \frac{P_0}{2} \cdot (1 + cos(\phi_B + \delta)) = \\ &= \frac{P_0}{2} \cdot (1 + \underbrace{cos(\phi_B)}_{0} + cos'(phi_B) \cdot \delta) = \\ &= \frac{P_0}{2} \cdot (1 - sin(phi_B) \cdot \delta) \end{aligned} \tag{14}$$

where $\phi_B \in \{+\frac{3}{2}\pi, +\frac{1}{2}\pi, -\frac{1}{2}\pi, -\frac{3}{2}\pi\}$.

$$
\begin{aligned}
e_{V_\pi}^{-\frac{1}{2}\pi}(n+1) &= P(n) - P(n+1) = \\
&= \frac{P_0}{2} \cdot \left(\frac{\delta_n + \delta_{n-2}}{2} - \frac{\delta_{n+1} + \delta_{n-1}}{2}\right) = \\
&= \frac{P_0}{4} \cdot (\underbrace{\delta_{n-2} - \delta_{n-1}}_{\text{last two inputs}} + \underbrace{\delta_n - \delta_{n+1}}_{\text{current two inputs}}) \\
e_{V_\pi}^{+\frac{1}{2}\pi}(n+3) &= P(n+2) - P(n+3) = \\
&= \frac{P_0}{2} \cdot \left(\frac{\delta_{n+2} + \delta_n}{2} - \frac{\delta_{n+3} + \delta_{n+1}}{2}\right) \\
&= \frac{P_0}{4} \cdot (\underbrace{\delta_n - \delta_{n+1}}_{\text{last two inputs}} + \underbrace{\delta_{n+2} - \delta_{n+3}}_{\text{current two inputs}})
\end{aligned} \tag{15}
$$

References

[Doe] *First and last LaTeX example.*, John Doe 50 B.C.